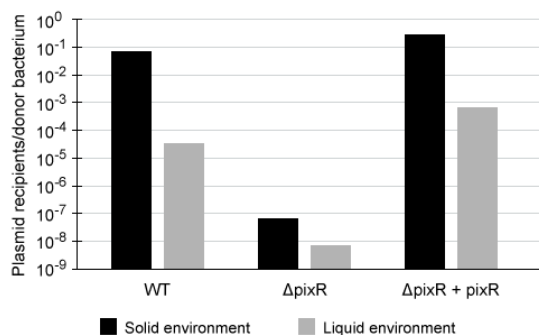


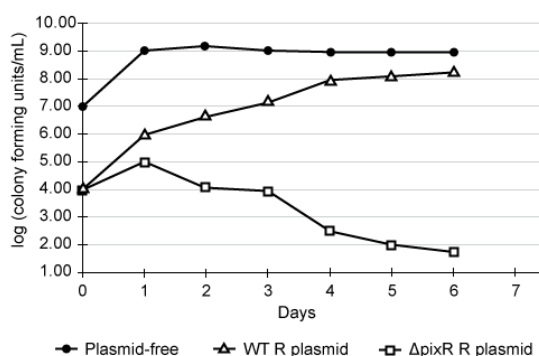
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**Figure 1** Bacteria receiving R plasmid from donor bacteria

In a second experiment, plasmid-free and plasmid-containing bacteria were coincubated over several days. Researchers monitored uptake of the plasmids from WT and  $\Delta$ *pixR* bacteria into plasmid-free recipient bacteria. The results are shown in Figure 2.



**Figure 2** Prevalence of plasmid-free and plasmid-containing bacteria during coincubation

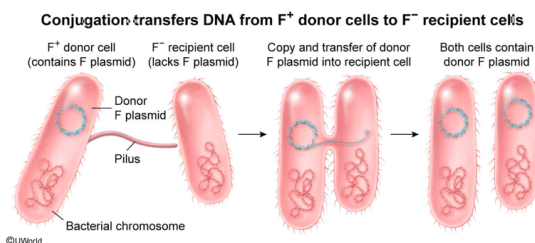
Which series shows the order in which DNA would pass by conjugation between two Gram-negative bacteria?

- ☐ A. Gram-negative bacterium with F plasmid → pilus → Gram-negative bacterium with F plasmid [6%]
- ✓ ☒ B. Gram-negative bacterium with F plasmid → pilus → Gram-negative bacterium lacking F plasmid [84%]
- ☐ C. Gram-negative bacterium lacking F plasmid → pilus → Gram-negative bacterium with F plasmid [6%]
- ☐ D. Gram-negative bacterium lacking F plasmid → pilus → Gram-negative bacterium lacking F plasmid [1%]

Correct

84% answered correctly

### Explanation:



**Conjugation** is a process whereby DNA is transferred between bacteria that are temporarily joined by a connecting tube called a pilus. The pilus pulls the cells together and forms a bridge through which the transferred DNA passes. The genetic information necessary for pilus formation is contained in a sequence of DNA called the fertility factor (F factor). Cells with the F factor are called F<sup>+</sup> cells when the F factor is contained within a **plasmid** and *Hfr* ("high frequency of recombination") cells when the F factor is contained within the bacterial chromosome.

The question asks for the order in which DNA is transferred between two Gram-negative bacteria during conjugation. **Transfer** of DNA during conjugation always **proceeds unidirectionally, from a bacterium possessing the F factor** (and hence able to form a pilus) **to a bacterium lacking the factor**. Therefore, in this case, transfer would proceed from the bacterium with the F plasmid, through the pilus, and to the bacterium lacking the F plasmid.

**(Choice A)** Only bacteria *lacking* the F factor receive DNA during conjugation; therefore, a Gram-negative bacterium with the F plasmid would *not* be a recipient cell during transfer of DNA by conjugation.

**(Choices C and D)** Only bacteria *possessing* the F factor donate DNA during conjugation; therefore, a Gram-negative bacterium lacking the F plasmid would *not* be a donor cell during conjugation.

**Educational objective:**

Conjugation is the transfer of DNA between bacteria through a tube formed by the pilus.

During conjugation, transfer occurs unidirectionally from cells with the F factor to cells lacking the F factor.

**Time spent:**0 sec

**Subject:**

Biology

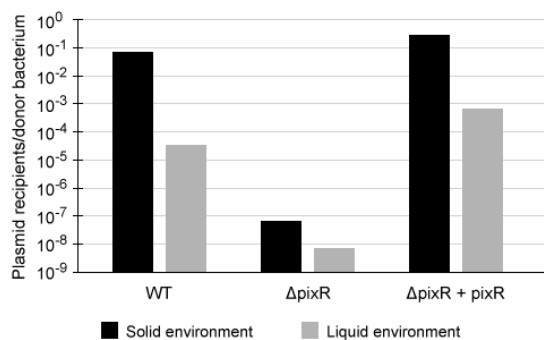
**QID:**403647

**System:**

Cellular Biology

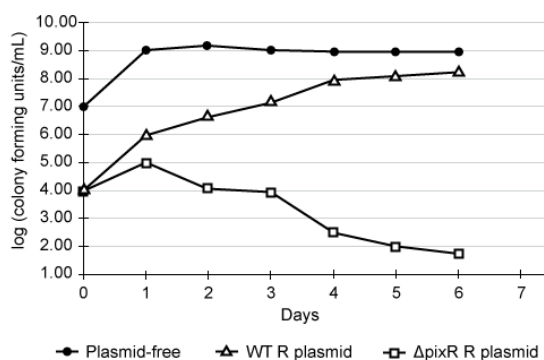
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**Figure 2** Prevalence of plasmid-free and plasmid-containing bacteria during coincubation

In Gram-negative bacteria, what is the maximum number of peptidoglycan layers through which crystal violet may pass during Gram staining?

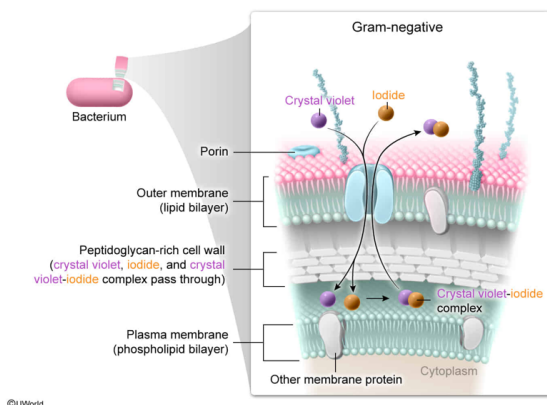
- ☐ A. 0 [26%]
- ✓ ☒ B. 1 [45%]
- ☐ C. 2 [24%]
- ☐ D. 3 [3%]

Correct

45% answered correctly

### Explanation:

Crystal violet crosses the peptidoglycan-rich cell wall during Gram staining



Bacteria typically have **cell walls** surrounding the **plasma membranes**. These cell walls are composed largely of **peptidoglycan**.

The passage describes the classification of bacteria as Gram-positive or Gram-negative based on the results of Gram staining, and the question asks for the maximum number of peptidoglycan layers through which crystal violet passes during Gram staining of Gram-negative bacteria.

As described in the passage, crystal violet and iodide can cross bacterial cell walls and the lipid membrane covering the cell wall of Gram-negative bacteria. Because peptidoglycans are found only in the cell wall, the **maximum number** of peptidoglycan layers **crystal violet may cross** is **one**.

**(Choice A)** Nearly all types of bacteria have a cell wall. Therefore, although a few types of Gram-negative bacteria lack a cell wall, the *maximum* number of peptidoglycan layers is greater than zero.

**(Choice C)** Gram-positive bacteria possess both a plasma membrane and a cell wall; however, of these two layers *only* the cell wall contains peptidoglycan.

**(Choice D)** Gram-negative bacteria possess a plasma membrane, a cell wall, and a second membrane layer outside the cell wall. However, of these three layers, *only* the cell

wall contains peptidoglycan.

**Educational objective:**

Bacteria are surrounded by a cell wall composed mainly of peptidoglycan.

**Time spent:**0 sec

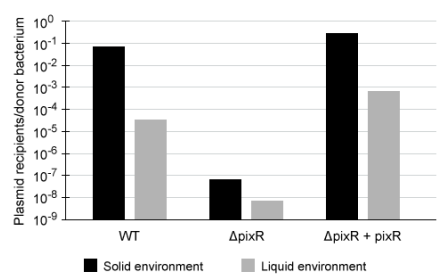
**Subject:**  
Biology

**QID:**403648

**System:**  
Cellular Biology

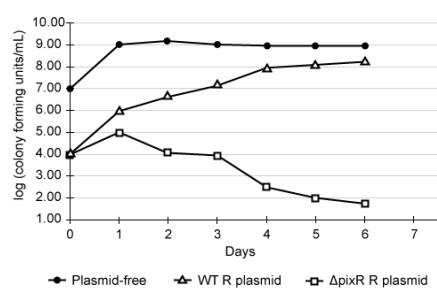
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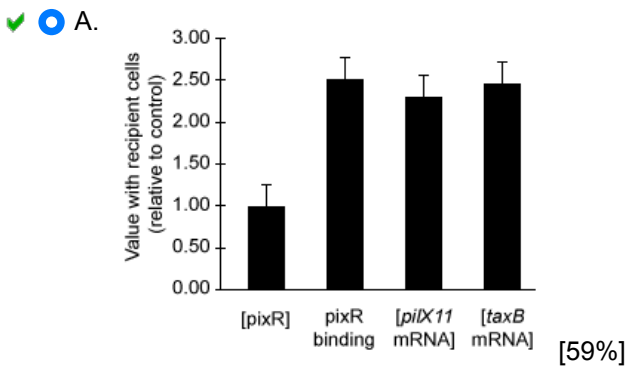
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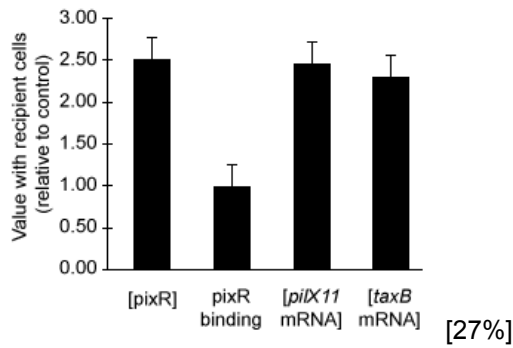


**Figure 2** Prevalence of plasmid-free and plasmid-containing bacteria during coincubation

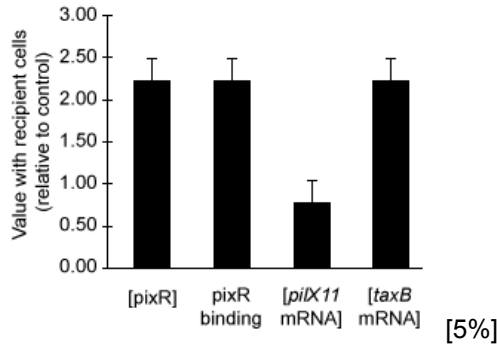
A positive control mechanism for bacterial transcription is one in which promoter binding by a protein stimulates transcription of the genes controlled by the promoter. Given this, which graph shows the expected results if the presence of recipient cells regulates transcription of the genes *pilX11* and *taxB* in the R plasmid via a positive control mechanism involving *pixR*?



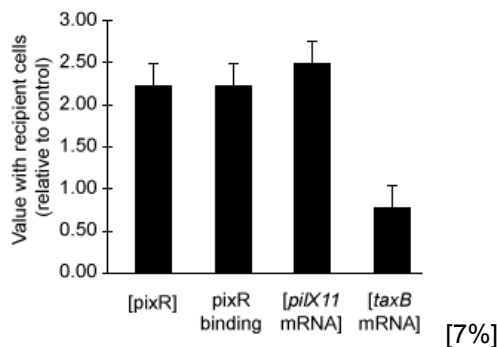
B.



C.



D.

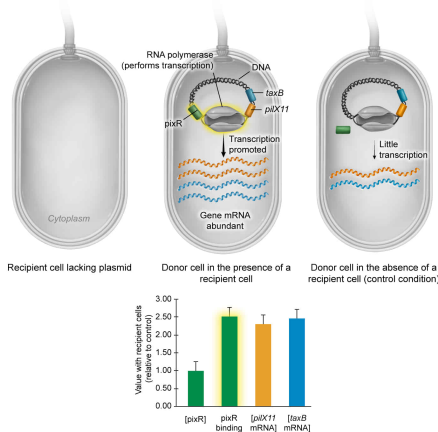


Correct

59% answered correctly

## Explanation:

Positive control of *pilX11* and *taxB* transcription by *pixR* binding



In bacterial cells, the **transcription** of genes that are necessary for the same process or function (eg, **lactose use**) are often under the control of a "switch" that increases or decreases their transcription at the same time. In such instances, binding of a specific transcriptional regulator protein (eg, **transcription factor**) at or near the RNA polymerase binding site serves to "flip the switch" to increase or decrease transcription of the genes under the control of the promoter.

The passage describes studies of the role of the protein *pixR* in the transcription of antibiotic resistance-promoting genes found along with *pixR* in a plasmid (ie, R plasmid). The question states that a positive control mechanism is one in which binding of an activating protein to a promoter causes increased transcription of the genes controlled by



the promoter. The question then asks for the answer choice graph depicting the expected results if *pixR* is part of a positive control mechanism activated by plasmid recipient cells and regulates transcription of the genes in the R plasmid.

In a **positive control mechanism**, the presence of recipient cells would increase *pixR* binding to the promoter for the antibiotic resistance genes, thereby stimulating transcription of the genes and leading to an increase in their mRNA levels. Note that in this mechanism, it is the extent of *pixR* binding that is important; *pixR* concentration is not relevant. Therefore, in the presence of recipient cells, relative ***pixR* binding** and the **concentrations** of *pilX11* and *taxB* mRNA would be **greater (ie, >1) than** under **control** conditions when the recipient cells were absent.

**(Choice B)** If *pixR* were part of a positive control mechanism activated by recipient cells, then the DNA-binding *activity* (not necessarily the concentration) of *pixR* would be increased compared to the control condition (ie, >1). Therefore, **this choice** does not depict a positive control mechanism as defined in the question.

**(Choices C and D)** In a positive control mechanism in which *pixR* regulates transcription of the *pilX11* and *taxB* genes, it would be expected that transcription of *both* genes would be increased compared to the control condition. Therefore, **these choices** do not depict a positive control mechanism as defined in the question.

#### Educational objective:

In a positive control mechanism of transcriptional regulation, protein binding to a promoter activates transcription of genes under the control of the promoter.

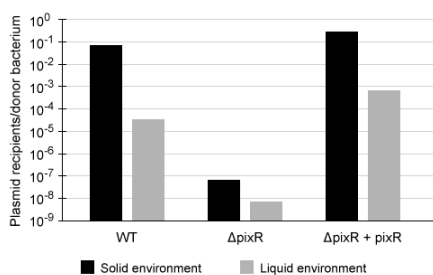
Time spent: 0 sec  
Subject:  
Biology

QID: 403649  
System:  
Cellular Biology



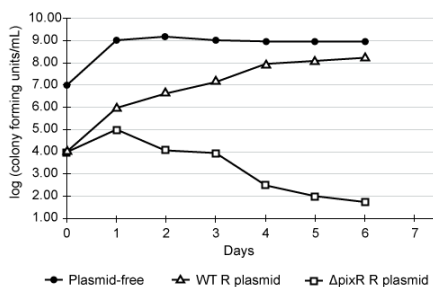
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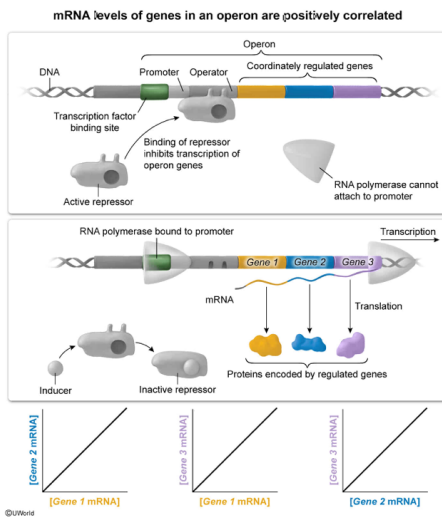
**Figure 2** Prevalence of plasmid-free and plasmid-containing bacteria during coincubation

Given that an operon is a set of genes regulated together with the same operator and promoter, which of the following relationships would be most likely to be found in genes that are part of the same operon?

- ✓ ☒ A. A positive correlation between mRNA transcript levels for the genes [63%]
- ☐ B. Negative correlations between each gene's mRNA transcript levels and the mRNA transcript level of an operon-regulating transcription factor [16%]
- ☐ C. Positive correlations between repressor binding to the operon's operator and the mRNA transcript levels for each gene [18%]
- ☐ D. Lack of correlation between transcription factor binding to the promoter and mRNA transcript levels for the genes [2%]

Correct  
62% answered correctly

**Explanation:**



An **operon** is composed of three elements:

- a promoter, or the upstream region of DNA to which RNA polymerase and other transcriptional modulators binds
- an operator, which is a nucleotide sequence, either within the promoter or between the promoter and the downstream genes regulated by the promoter, to which a repressor binds
- the genes controlled by the operon

The question asks for the relationship most likely to be found for genes that are part of the same operon. If part of the same operon, **mRNA transcript levels of the genes should be positively correlated**. In other words, if the mRNA level of one gene in the operon increases or decreases, the mRNA levels for other genes in the operon should also increase or decrease, respectively.

**(Choice B)** Transcription factors (eg, repressors) bind to the promoter region of the gene(s) they regulate.

Accordingly, *proteins* that regulate the switching of an operon on or off bind to the promoter region for genes in the operon. Although binding of some transcription factors (eg, a repressor *protein*) should be negatively correlated with each gene's mRNA transcript level, the *mRNA level* of transcription factors regulating an operon may or may

not be correlated with each gene's mRNA transcript level.

**(Choice C)** Repressor binding to the operons' operator is expected to decrease transcription of genes in the operon. Therefore, negative (*not* positive) correlations between repressor binding and the mRNA transcript levels for each gene would be expected.

**(Choice D)** Transcription factor binding to a promoter either promotes or inhibits transcription of genes regulated by the promoter. Therefore, one would expect that mRNA levels of genes in an operon would be positively or negatively correlated with transcription factor binding to the promoter for the operon.

### Educational objective:

Operons consist of a set of genes whose expression is correlated due to regulation by the same promoter and operator.

Time spent: 0 sec

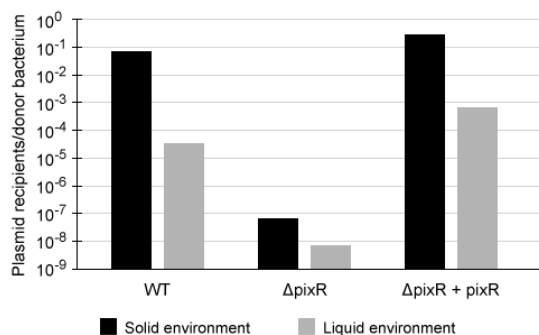
Subject:  
Biology

QID: 403650

System:  
Cellular Biology

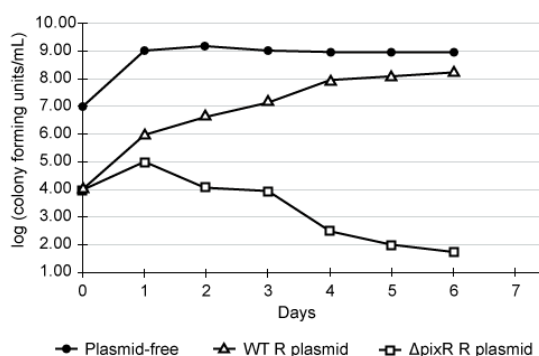
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**Figure 2** Prevalence of plasmid-free and plasmid-containing bacteria during coincubation

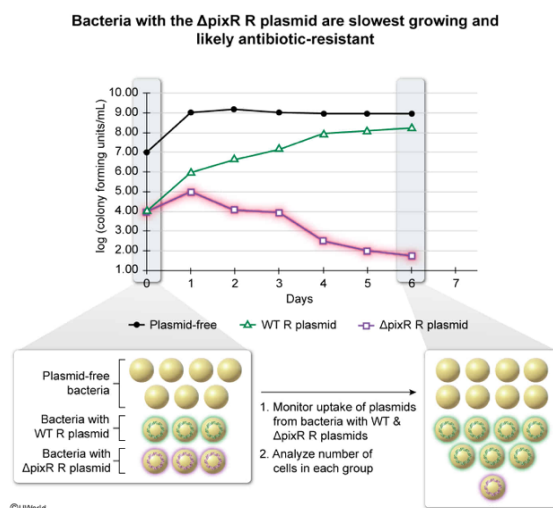
Which of the following best describes the likely phenotype of the bacterial cells possessing the  $\Delta$ pixR R plasmid in Figure 2?

- ☐ A. More antibiotic-susceptible and slower growing than the cells possessing wild-type R plasmids [72%]
- ☐ B. More antibiotic-susceptible and faster growing than the cells possessing wild-type R plasmids [4%]
- ☒ C. More antibiotic-resistant and slower growing than the cells lacking plasmids [16%]
- ☐ D. More antibiotic-resistant and faster growing than the cells lacking plasmids [6%]

Correct

15% answered correctly

### Explanation:



Because they do not engage in sexual reproduction, bacteria do not **increase** their genetic variation through **fertilization**. Nevertheless, bacteria and other prokaryotes possess an impressive ability to transmit genetic information by **other means**, thereby enabling rapid genetic adaptation to changing environmental conditions. For example, transmission of an antibiotic resistance gene to antibiotic susceptible bacteria via plasmids can allow the bacteria to survive and grow during treatment with the antibiotic.

The passage describes an experiment in which plasmid-free and plasmid-containing bacteria were incubated together over several days. Uptake of the WT and  $\Delta$ pixR plasmids from WT and  $\Delta$ pixR bacteria into plasmid-free recipient bacteria was monitored by the researchers. Figure 2 in the passage shows the amounts of bacteria in which the R plasmids are either present or absent, and the question asks for the statement that best describes the phenotype of the bacterial cells possessing the mutant plasmid in which *pixR* is deleted (ie,  $\Delta$ pixR).

The number of bacteria with the  $\Delta$ pixR R plasmid increases briefly before exhibiting a steady decline over time. Furthermore, after the initial time point, bacterial numbers in the group with the  $\Delta$ pixR R plasmid are always lower than bacterial numbers in the other two groups. As shown in Figure 1, bacteria with the  $\Delta$ pixR R plasmid (ie, lacking *pixR*) are less capable of passing the R plasmid to recipient bacteria. However, these bacteria still possess the

other genes in the R plasmid. Therefore, **bacteria with the  $\Delta$ pixR R plasmid are slower-growing** than the cells lacking either plasmid (Figure 2) **and are likely antibiotic-resistant.**

**(Choices A and B)** Bacteria with the  $\Delta$ pixR R plasmid and bacteria with the WT R plasmid are likely *equally* antibiotic-resistant because they both contain an antibiotic resistance gene.

**(Choice D)** The number of bacteria with the  $\Delta$ pixR R plasmid declines for most of the experiment and is always lower than the number of plasmid-free bacteria. The number of plasmid-free bacteria remains stable through most of the experiment after initially increasing at a faster absolute and relative rate than the bacteria with the  $\Delta$ pixR R plasmid (ie, lacking pixR). Therefore, the bacteria lacking pixR are *slower* growing than the plasmid-free bacteria.

**Educational objective:**

Plasmids carrying antibiotic resistance genes can transmit antibiotic resistance to recipient cells.

Time spent:0 sec

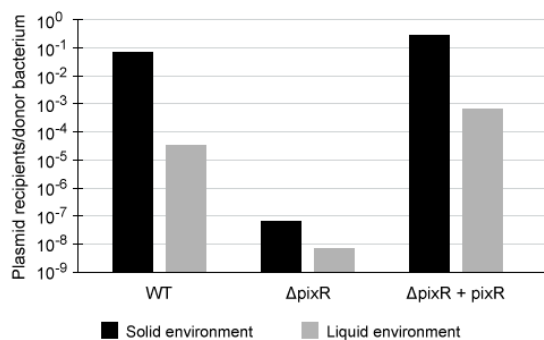
Subject:  
Biology

QID:403651

System:  
Cellular Biology

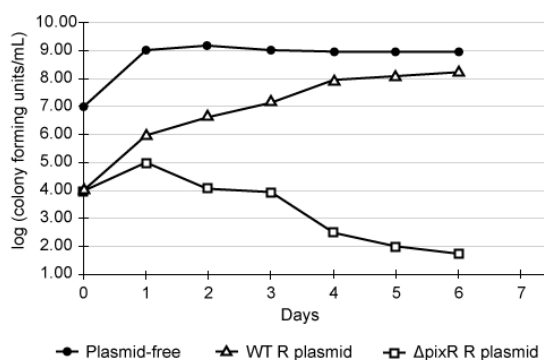
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**Figure 2** Prevalence of plasmid-free and plasmid-containing bacteria during coincubation



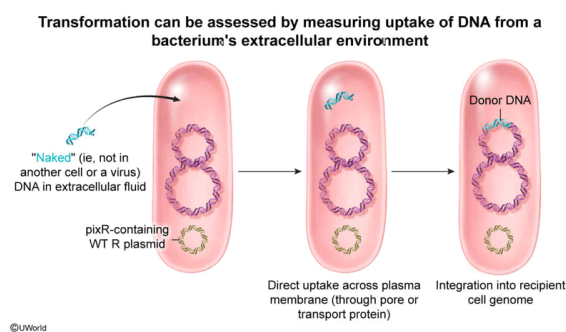
Which experiment comparing plasmid-free bacteria and bacteria with the WT R plasmid could the researchers use to investigate whether *pixR* regulates bacterial transformation?

- ☐ A. Assessing uptake of extracellular RNA [11%]
- ✓ ☒ B. Assessing uptake of extracellular DNA [67%]
- ☐ C. Assessing uptake of DNA from DNA viruses [11%]
- ☐ D. Assessing uptake of RNA from RNA viruses [9%]

Correct

66% answered correctly

### Explanation:



Genetic variation in bacteria is augmented by recombination, the mixing of DNA from different sources. This recombination occurs by three general mechanisms: (1) **transduction**, (2) conjugation, and (3) transformation.

**Transformation** is the simple **uptake of DNA** (either a linear fragment or a plasmid) from a recipient bacterium's extracellular fluid through membrane pores or transport proteins. Transformation does not require the joining of a recipient bacterium to a virus or to a donor bacterium.

The question asks for the experiment comparing plasmid-free bacteria and bacteria with the WT R plasmid that the researchers could use to investigate whether *pixR* regulates bacterial transformation. Of the choices, only **assessment of extracellular DNA uptake** represents the source (ie, extracellular fluid) and means (transport in the absence of a donor bacterium or virus) by which DNA is transferred to a recipient cell during transformation.

**(Choice A)** Transformation involves uptake of DNA (*not* RNA) from a bacterium's extracellular environment.

**(Choices C and D)** Transformation involves uptake of DNA (*not* RNA) from a bacterium's extracellular environment (*not* from a virus).

### Educational objective:

During transformation, DNA is taken up by a cell from the extracellular fluid surrounding the cell.

**Time spent:**0 sec  
**Subject:**  
Biology

**QID:**403652  
**System:**  
Cellular Biology